Optimized Integration of Periprandial Transition Profiles and Enteropancreatic Patterns with Glycemic Subclassification

A Deterministic Classification Framework for Cross-Sectional and Post-Intervention Characterization of Enteropancreatic Physiology

Technical Methods Document

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Classification framework for analyte-level Periprandial Transition Profiles (PTPs) and rule-based Integrated Enteropancreatic Patterns (IEPs, Types I–V) based on Functional Principal Component Analysis of periprandial hormone dynamics, standardized against a Lean-Healthy reference distribution.

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1. Introduction and Rationale

The characterization of enteroendocrine and pancreatic hormone dynamics in response to nutrient stimuli represents a fundamental challenge in metabolic endocrinology. Traditional approaches to classifying postprandial hormone responses—relying on scalar summary measures such as area under the curve (AUC) or peak concentration values—fail to capture the rich temporal structure embedded in periprandial time-series data (Ramsay & Silverman, 2005; Müller, 2005). The advent of Functional Data Analysis (FDA) and, specifically, Functional Principal Component Analysis (FPCA), provides a mathematically rigorous framework for decomposing hormone response curves into interpretable modes of variation: global amplitude (PC1), early-versus-late temporal redistribution (PC2), and biphasic or higher-order shape features (PC3) (Yao et al., 2005; Wang et al., 2016).

This document presents an optimized, deterministic classification framework that integrates analyte-level Periprandial Transition Profiles (PTPs) into Integrated Enteropancreatic Patterns (IEPs, Types I–V) with glycemic subclassification. The framework is designed for application to cross-sectional disease characterization (primary PTPs) and post-intervention evaluation (secondary PTPs), with explicit reference to a Lean-Healthy control distribution. The classification system addresses several critical methodological refinements:

First, **the term "secretory capacity" is replaced throughout by "physiologic periprandial responsiveness,"** a term that applies uniformly to secretion-dominant analytes (GIP, GLP-1, insulin), suppression-dominant signals (ghrelin), and glucose, which is not a secretory hormone but rather a glycemic regulation signal (Nauck & Müller, 2023; Holst, 2007).

Second, **the primary classification coordinates are strictly defined** as the basal concentration, PC1 (global amplitude), ξ_2/PC2 (early-versus-late temporal redistribution), and ξ_3/PC3 (biphasic or higher-order shape). The secondary orthogonal shape-balance descriptor derived from ξ_2 and ξ_3 remains supplementary and must not replace these primary coordinates in the classification rule (Ramsay & Silverman, 2005).

Third, **weight loss is not used as a hard gate for secondary PTP labels.** Post-intervention classification depends on where the post-intervention profile lies relative to the Lean-Healthy reference and on the direction of change relative to the pre-intervention state. Weight loss is reported as a contextual modifier only (Mingrone et al., 2012; Schauer et al., 2017).

2. General Notation and Analyte Groupings

2.1 Standardized Analyte-Specific Classification Vector

For analyte h and record i, the classification uses the standardized analyte-specific vector:

$$\mathbf{Z_ih = [\ z(B_ih),\ \ z(\xi_ih1),\ \ z(\xi_ih2),\ \ z(\xi_ih3)\ ]}$$

Where:

B_ih = basal (fasting) concentration for analyte h in record i

ξ_ih1 = PC1 score (global amplitude component)

ξ_ih2 = PC2 score (early-versus-late temporal redistribution)

ξ_ih3 = PC3 score (biphasic or higher-order temporal shape)

The secondary orthogonal shape-balance descriptor derived from ξ_2 and ξ_3 may be reported as a supplementary summary of temporal-shape balance but must not replace ξ_2 and ξ_3 in the primary classification rule.

2.2 Analyte Groupings

Signal Type	Analytes	Physiological Rationale
Secretion-dominant	GIP, GLP-1, PYY, Insulin	Postprandial secretory response is the primary dynamic; classification tracks amplitude and temporal redistribution of secretion
Suppression-dominant	Ghrelin, Glucagon	Postprandial suppression is the primary dynamic; classification tracks the depth and temporal profile of suppression (Engström et al., 2007; Samat et al., 2013)
Glycemic regulation signal	Glucose	Not a secretory hormone; classification tracks glycemic excursion dynamics, redistribution, and regulation rather than secretory capacity

2.3 Preferred Statistical Implementation

For temporal-shape classification, a bivariate reference region for (ξ_2, ξ_3) is preferred, ideally constructed as a Lean-Healthy reference ellipse (e.g., 95% tolerance ellipse from the Lean-Healthy cohort). If a multivariate reference region is not available, a percentile-based fallback rule for ξ_2 and ξ_3 separately should be applied. The operational percentile thresholds specified below serve as this fallback (Ramsay & Silverman, 2005; Wang et al., 2016).

3. Primary Enteropancreatic Periprandial Transition Profiles

Primary PTPs provide the cross-sectional characterization of metabolic disease states by comparing disease cohorts against a Lean-Healthy reference distribution. Six classes are defined, spanning from reference-concordant physiology through lower-tail impairment to upper-tail dysregulation.

3.1. Preserved

Definition: *The periprandial transition is reference-concordant.*

Interpretation: Preserved physiologic periprandial responsiveness, without clear evidence of attenuated, exaggerated, dysregulated, or pathologically shifted transition.

Preferred statistical criteria: Assign Preserved when: basal concentration remains within the Lean-Healthy central reference band; PC1 remains within the central reference band; ξ_2 and ξ_3 remain within the reference-compatible temporal-shape region; and no primary coordinate crosses a lower-tail or upper-tail abnormality threshold.

Operational percentile fallback: Basal: 25th–75th percentile; PC1: 25th–75th percentile; ξ_2, ξ_3: compatible with the central reference region.

Physiological note: *Secretion-dominant analytes: preserved secretory responsiveness. Ghrelin: preserved postprandial suppression. Glucose: preserved glycemic dynamic regulation.*

3.2. Borderline Impaired

Definition: *The periprandial transition shows a mild lower-tail deviation relative to Lean-Healthy.*

Interpretation: Mild reduction in physiologic periprandial responsiveness, consistent with early impairment, reduced early-phase recruitment, deficient suppression, or attenuated coordinated response.

Preferred statistical criteria: Assign Borderline Impaired when: at least one primary coordinate shows a lower-tail deviation in the mild abnormal range; there is no severe lower-tail shift; and there is no competing upper-tail dysregulation that would justify an Altered label.

Operational percentile fallback: Basal and/or PC1: 10th–25th percentile; ξ_2 and/or ξ_3: mild deviation toward impaired temporal structure; No primary coordinate <5th percentile.

Physiological note: *Avoid describing this automatically as hypoglycemia. For glucose, this more often indicates attenuated or poorly distributed glycemic excursion, not necessarily clinically meaningful hypoglycemia.*

3.3. Impaired

Definition: *The periprandial transition shows a moderate lower-tail deviation relative to Lean-Healthy.*

Interpretation: Moderately reduced physiologic periprandial responsiveness, compatible with persistent dysfunction in secretion, suppression, early-phase recruitment, or coordinated postprandial control.

Preferred statistical criteria: Assign Impaired when: at least one primary coordinate falls in the moderate lower-tail abnormal range; the profile is more abnormal than Borderline Impaired but does not meet Blunted criteria.

Operational percentile fallback: Basal and/or PC1: 5th–10th percentile; ξ_2 and/or ξ_3: moderate deviation toward impaired temporal structure; No primary coordinate <5th percentile.

3.4. Blunted

Definition: *The periprandial transition is severely attenuated relative to Lean-Healthy.*

Interpretation: Severely reduced physiologic periprandial responsiveness, consistent with marked attenuation of secretion, suppression, or glycemic dynamic reactivity.

Preferred statistical criteria: Assign Blunted when: one or more primary coordinates fall in the extreme lower tail; and the temporal-shape structure is markedly flattened, delayed, or otherwise severely attenuated.

Operational percentile fallback: Basal and/or PC1: <5th percentile; ξ_2 and/or ξ_3: marked deviation consistent with attenuated transition.

Physiological note: *For glucose, use this label only when the dynamic excursion/regulation pattern is truly severely attenuated relative to reference; do not equate it automatically with severe hypoglycemia.*

3.5. Borderline Altered

Definition: *The periprandial transition shows a mild upper-tail deviation in a pathophysiologically relevant direction.*

Interpretation: Mildly abnormal physiologic periprandial responsiveness, compatible with excessive excursion, inappropriate persistence, impaired suppression, or early dysregulation rather than preserved enhancement.

Preferred statistical criteria: Assign Borderline Altered when: at least one pathophysiologically relevant primary coordinate falls in the mild upper-tail abnormal range; the pattern does not yet reach the severity required for Altered.

Operational percentile fallback: Pathophysiologically relevant Basal and/or PC1: 75th–95th percentile; ξ_2 and/or ξ_3: upper-tail deviation consistent with dysregulated temporal redistribution or exaggerated structure.

Physiological note: *In primary disease characterization, this label is most relevant for: GIP, insulin, glucose, and contextually ghrelin and for GLP-1 without hipoglicemia..*

3.6. Altered

Definition: *The periprandial transition is clearly and pathophysiologically shifted above the Lean-Healthy reference.*

Interpretation: Markedly abnormal physiologic periprandial responsiveness, consistent with hyperresponsive, dysregulated, inappropriately persistent, or maladaptively elevated postprandial dynamics.

Preferred statistical criteria: Assign Altered when: at least one pathophysiologically relevant primary coordinate lies in the extreme upper tail; and the profile is incompatible with reference-concordant physiology.

Operational percentile fallback: Pathophysiologically relevant Basal and/or PC1: ≥95th percentile; ξ_2 and/or ξ_3: upper-tail deviation consistent with dysregulated temporal structure.

Physiological note: *Use this label primarily for: excessive GLP-1 without hipoglicemia, GIP or dysregulated GIP dynamics; hyperinsulinemic patterns interpreted jointly with glucose; hyperglycemic excursion or prolonged late redistribution; and context-specific ghrelin excess.*

3.7. Statistical Note for Primary PTPs

For primary disease characterization, upper-tail classes should be applied only when excess is pathophysiologically meaningful for that analyte in that context. Do not automatically classify high GLP-1 as "Altered" in cross-sectional disease data unless there is clear biological justification. The incretin effect in type 2 diabetes is known to be impaired for GIP but relatively preserved for GLP-1 (Nauck & Müller, 2023), and elevated GLP-1 concentrations post-bariatric surgery represent physiologically coherent enhancement rather than dysregulation (Laferrère, 2011; Pournaras et al., 2010).

4. Secondary Enteropancreatic Periprandial Transition Profiles

Secondary PTPs classify post-intervention profiles following caloric restriction, bariatric surgery, pharmacotherapy, or other therapeutic interventions. Classification depends on: (a) the post-intervention position relative to the Lean-Healthy reference, and (b) the direction of change relative to the pre-intervention state. Weight loss is treated as a contextual modifier, not a hard gate.

4.1. Preserved (Secondary)

Definition: *A reference-concordant profile remains reference-concordant after intervention.*

Interpretation: Preserved physiologic periprandial responsiveness is maintained post-intervention.

Criteria: Assign Preserved when: the post-intervention profile is reference-concordant; the pre-intervention profile was already preserved; no meaningful deterioration or exaggerated shift is present post-intervention.

Operational percentile fallback: Post-intervention Basal, PC1: 25th–75th percentile; ξ_2 and ξ_3: reference-compatible temporal-shape structure.

4.2. Recovered

Definition: *A previously abnormal profile becomes reference-concordant after intervention.*

Interpretation: Restoration of physiologic periprandial responsiveness toward the Lean-Healthy reference.

Criteria: Assign Recovered when: the pre-intervention profile was Borderline Impaired, Impaired, Blunted, Borderline Altered, or Altered; and the post-intervention profile shifts into the reference-concordant region.

Operational percentile fallback: Post-intervention Basal, PC1: 25th–75th percentile; ξ_2 and ξ_3: reference-compatible. Explicit improvement from a previously abnormal state required.

4.3. Borderline Impaired (Secondary)

Definition: *A mild lower-tail abnormality persists or emerges after intervention.*

Interpretation: Residual mild reduction in physiologic periprandial responsiveness despite treatment.

Criteria: Assign Borderline Impaired when: post-intervention lower-tail abnormality remains in the mild range.

Operational percentile fallback: Post-intervention Basal and/or PC1: 10th–25th percentile; ξ_2 and/or ξ_3: mild lower-tail deviation.

4.4. Impaired (Secondary)

Definition: *A moderate lower-tail abnormality persists after intervention.*

Interpretation: Persistent moderate reduction in physiologic periprandial responsiveness despite treatment.

Criteria: Assign Impaired when: post-intervention moderate lower-tail deviation remains; the profile remains clearly abnormal but not severely attenuated.

Operational percentile fallback: Post-intervention Basal and/or PC1: 5th–10th percentile; ξ_2 and/or ξ_3: moderate lower-tail deviation.

4.5. Blunted (Secondary)

Definition: *A severe lower-tail abnormality persists after intervention.*

Interpretation: Markedly attenuated physiologic periprandial responsiveness despite treatment.

Criteria: Assign Blunted when: post-intervention one or more primary axes remain <5th percentile; and severe attenuation persists despite intervention.

Operational percentile fallback: Post-intervention Basal and/or PC1: <5th percentile; ξ_2 and/or ξ_3: marked lower-tail deviation.

4.6. Borderline Altered (Secondary)

Definition: *A mild upper-tail dysregulated pattern persists or emerges after intervention.*

Interpretation: Residual or emerging mild pathophysiologic exaggeration or dysregulation of physiologic periprandial responsiveness after treatment.

Criteria: Assign Borderline Altered when: post-intervention upper-tail deviation is present in the mild abnormal range; the profile is not best interpreted as physiologically coherent enhancement.

Operational percentile fallback: Post-intervention pathophysiologically relevant Basal and/or PC1: 75th–95th percentile; ξ_2 and/or ξ_3: upper-tail dysregulated deviation.

4.7. Altered (Secondary)

Definition: *A marked upper-tail dysregulated pattern persists or emerges after intervention.*

Interpretation: Persistent or newly emergent pathophysiologic excess, mismatch, or dysregulation of physiologic periprandial responsiveness after treatment.

Criteria: Assign Altered when: post-intervention upper-tail deviation reaches the extreme abnormal range; the pattern is incompatible with physiologically coherent recovery or enhancement.

Operational percentile fallback: Post-intervention pathophysiologically relevant Basal and/or PC1: ≥95th percentile; ξ_2 and/or ξ_3: marked upper-tail dysregulated deviation.

4.8. Borderline Enhanced

Definition: *A post-intervention supraphysiologic increase is present but remains physiologically coherent.*

Interpretation: Mildly supraphysiologic physiologic periprandial responsiveness, typically seen after effective intervention, especially for incretin and satiety axes.

Criteria: Assign Borderline Enhanced when: the post-intervention profile lies in the mild upper tail; the increase is physiologically coherent rather than dysregulated; glucose remains preserved or recovered; and the analyte context supports enhancement rather than alteration.

Operational percentile fallback: Post-intervention PC1: 75th–95th percentile; Reference-compatible or adaptive ξ_2/ξ_3 behavior; Preserved or recovered glucose context required when relevant.

Physiological note: *This label is most appropriate for: GLP-1 and contextually insulin, only when glucose is preserved or recovered. Post-RYGB GLP-1 amplification with preserved glycemia exemplifies this category (Pournaras et al., 2010; Bojsen-Møller et al., 2015).*

4.9. Enhanced

Definition: *A marked post-intervention supraphysiologic increase is present and remains physiologically coherent.*

Interpretation: Markedly increased but physiologically coherent physiologic periprandial responsiveness, often compatible with post-bariatric incretin/satiety amplification.

Criteria: Assign Enhanced when: the post-intervention profile lies in the extreme upper tail; the increase is physiologically coherent rather than dysregulated; glucose is preserved or recovered; and the analyte context supports enhancement.

Operational percentile fallback: Post-intervention PC1: ≥95th percentile; Reference-compatible or adaptively shifted ξ_2/ξ_3; Preserved or recovered glucose context required when relevant.

Physiological note: *Reserve Enhanced mainly for: GLP-1 and carefully selected insulin responses with preserved glycemic regulation. Do not use "Enhanced" if the accompanying glucose pattern is altered or dysglycemic. In that setting, classify the profile as Altered, not Enhanced.*

5. Summary of Percentile-Based Operational Thresholds

5.1 Primary PTP Thresholds (Cross-Sectional)

PTP Class	Tail	Basal / PC1	ξ_2 / ξ_3	Interpretation
Preserved	Central	25th–75th	Central reference	Reference-concordant
Borderline Impaired	Lower	10th–25th	Mild lower deviation	Mild impairment
Impaired	Lower	5th–10th	Moderate lower deviation	Moderate impairment
Blunted	Lower	<5th	Marked lower deviation	Severe attenuation
Borderline Altered	Upper	75th–95th	Mild upper deviation	Mild dysregulation
Altered	Upper	≥95th	Marked upper deviation	Marked dysregulation

5.2 Secondary PTP Thresholds (Post-Intervention)

Secondary PTPs use the same percentile boundaries as primary PTPs, with the addition of two post-intervention–specific classes:

PTP Class	Post-Intervention Criteria	Pre-Intervention Requirement	Key Condition
Preserved	Basal/PC1: 25th–75th	Was already Preserved	Maintained
Recovered	Basal/PC1: 25th–75th	Was abnormal	Improved to reference
Borderline Impaired	Basal/PC1: 10th–25th	Any	Residual mild impairment
Impaired	Basal/PC1: 5th–10th	Any	Persistent moderate impairment
Blunted	Basal/PC1: <5th	Any	Persistent severe attenuation
Borderline Altered	Basal/PC1: 75th–95th	Any	Residual mild dysregulation
Altered	Basal/PC1: ≥95th	Any	Persistent marked dysregulation
Borderline Enhanced	PC1: 75th–95th	Any	Coherent + glucose preserved/recovered
Enhanced	PC1: ≥95th	Any	Coherent + glucose preserved/recovered

6. Construction of Integrated Enteropancreatic Patterns

Integrated Enteropancreatic Pattern (IEP) typing was performed at the cohort-time-arm level after deterministic assignment of analyte-level PTPs. Integrated typing was treated as a rule-based ecological classification rather than an unsupervised phenotype-discovery procedure.

6.1 Classifiability Rule

An integrated pattern was assigned only when:

(a) glucose PTP was classifiable; **(b)** at least two non-glucose analyte-level PTPs were classifiable; and **(c)** the classifiable non-glucose set included at least one pancreatic effector signal (insulin) and at least one gut-hormone signal (ghrelin, GIP, GLP-1).

Cohort-time arms not meeting these criteria were classified as "not integrable" and were excluded from Type I–V assignment. Missing analytes were not imputed.

6.2 Non-Glucose PTP State Groups

Group	PTP States Included	Direction
R	Preserved, Recovered	Reference-concordant
L1	Borderline Impaired	Mild lower tail
L2	Impaired, Blunted	Moderate–severe lower tail
U1	Borderline Enhanced	Mild upper (coherent)
U2	Enhanced	Marked upper (coherent)
D1	Borderline Altered	Mild upper (dysregulated)
D2	Altered	Marked upper (dysregulated)

6.3 Glucose Subtype Groups

Subtype	Glucose PTP States	Interpretation
a	Preserved, Recovered	Glycemic regulation preserved or restored
b	Borderline Impaired, Impaired, Blunted	Glycemic impairment / attenuation
c	Borderline Altered, Altered	Glycemic dysregulation / excess

Glucose was treated as a glycemic regulation signal rather than as an enteropancreatic hormone for integrated pattern assignment.

6.4 Deterministic Precedence Hierarchy

Integrated pattern assignment followed the deterministic precedence hierarchy below. The rule is evaluated top-to-bottom; the first matching condition determines the type.

Priority	Condition	Assigned Type
1 (highest)	Any non-glucose analyte is Altered	Type IV.II
2	Any non-glucose analyte is Borderline Altered	Type IV.I
3a	Any non-glucose analyte is Enhanced AND glucose = a AND no L1/L2	Type II.II
3b	Any non-glucose analyte is Enhanced (otherwise)	Type V.II
4a	Any non-glucose analyte is Borderline Enhanced AND glucose = a AND no L1/L2	Type II.I
4b	Any non-glucose analyte is Borderline Enhanced (otherwise)	Type V.I
5	Any non-glucose analyte is Impaired or Blunted	Type III.II
6	Any non-glucose analyte is Borderline Impaired	Type III.I
7	All classifiable non-glucose analytes are Preserved	Type I.I
8 (lowest)	At least one Recovered, rest Preserved/Recovered	Type I.II

7. Integrated Enteropancreatic Pattern Type Definitions

7.1 Type I: Physiological Enteropancreatic–Metabolic State

Type I.I: All classifiable non-glucose PTPs are Preserved and glucose subtype = a.

Type I.II: At least one classifiable non-glucose PTP is Recovered, all remaining classifiable non-glucose PTPs are Preserved or Recovered, and glucose subtype = a. No classifiable non-glucose PTP may be in L1, L2, U1, U2, D1, or D2.

Interpretation: *Reference-concordant or restored physiologic periprandial responsiveness, without evidence of infra-physiological, dysphysiological, or maladaptively supra-physiological non-glucose signaling.*

7.2 Type II: Supra-Physiological but Metabolically Coherent State

Type II.I: At least one classifiable non-glucose PTP is Borderline Enhanced, all remaining classifiable non-glucose PTPs are Preserved or Recovered, glucose subtype = a, and no classifiable non-glucose PTP is in L1, L2, D1, or D2.

Type II.II: At least one classifiable non-glucose PTP is Enhanced, all remaining classifiable non-glucose PTPs are Preserved or Recovered, glucose subtype = a, and no classifiable non-glucose PTP is in L1, L2, D1, or D2.

Interpretation: *Supra-reference enteropancreatic responsiveness that remains physiologically coherent because glycemic regulation is preserved or recovered. This pattern is particularly relevant for post-bariatric incretin/satiety amplification (Pournaras et al., 2010; Laferrère, 2011, 2016).*

7.3 Type III: Infra-Physiological Enteropancreatic State

Type III.I: At least one classifiable non-glucose PTP is Borderline Impaired, no classifiable non-glucose PTP is in L2, U1, U2, D1, or D2.

Type III.II: At least one classifiable non-glucose PTP is Impaired or Blunted, and no classifiable non-glucose PTP is in U1, U2, D1, or D2.

Interpretation: *Infra-physiological enteropancreatic responsiveness characterized by reduced secretion, reduced suppression, impaired redistribution, or attenuated functional dynamics, with glycemic status captured separately by subtype. Subtypes: III.Ia/III.IIa (glucose a), III.Ib/III.IIb (glucose b), III.Ic/III.IIc (glucose c).*

7.4 Type IV: Dysphysiological Enteropancreatic State

Type IV.I: At least one classifiable non-glucose PTP is Borderline Altered.

Type IV.II: At least one classifiable non-glucose PTP is Altered.

Interpretation: *Dysphysiological enteropancreatic organization driven by pathologically shifted or maladaptive non-glucose responses, irrespective of whether other analytes are preserved, reduced, or enhanced. Subtypes: IV.Ia/IV.IIa (glucose a), IV.Ib/IV.IIb (glucose b), IV.Ic/IV.IIc (glucose c).*

7.5 Type V: Supra-Physiological State with Residual Metabolic Dysregulation

Type V.I: At least one classifiable non-glucose PTP is Borderline Enhanced, no D1/D2, and the arm does not satisfy Type II.I because glucose subtype is b/c and/or one or more non-glucose analytes are in L1 or L2.

Type V.II: At least one classifiable non-glucose PTP is Enhanced, no D1/D2, and the arm does not satisfy Type II.II because glucose subtype is b/c and/or one or more non-glucose analytes are in L1 or L2.

Interpretation: *Supra-reference enteropancreatic responsiveness occurring in the presence of residual glycemic abnormality and/or mixed lower-tail non-glucose abnormalities. Physiologically distinct from Type II because the supra-physiological signal is not metabolically clean. Subtypes: V.Ia/V.IIa (glucose a), V.Ib/V.IIb (glucose b), V.Ic/V.IIc (glucose c).*

8. Statistical and Reporting Rules

Rule 1. Integrated pattern typing should be reported only in the classifiable subset.

Rule 2. Exact frequency estimates must be interpreted as ecological cohort-time-arm counts, not participant-level prevalence.

Rule 3. Primary-corpus Type I–V frequencies must be reported separately from corroborative source-reported pattern information.

Rule 4. The secondary orthogonal shape-balance descriptor derived from ξ_2 and ξ_3 may be reported descriptively but must not replace ξ_2 and ξ_3 in integrated typing.

Rule 5. When contradictory non-glucose PTP states are present, the deterministic precedence hierarchy (Section 6.4) determines the final integrated type.

9. Physiological Rationale and Evidence Base

The classification framework presented here is grounded in several converging lines of evidence from enteroendocrine physiology, incretin biology, and bariatric-metabolic surgery outcomes.

9.1 Incretin Hormone Dynamics and the Incretin Effect

Based on articles retrieved from PubMed, the incretin effect—the greater insulin secretory response following oral glucose compared with isoglycemic intravenous glucose—is markedly reduced or absent in type 2 diabetes (Nauck & Müller, 2023). This reduction appears primarily driven by impaired insulinotropic action of GIP, while GLP-1 retains substantial potency. After Roux-en-Y gastric bypass (RYGB), postprandial GLP-1 concentrations increase three- to five-fold, and this enhancement is sustained over years (Laferrère, 2011; Laferrère, 2016). The exaggerated GLP-1 response post-RYGB is physiologically coherent when accompanied by preserved glycemic regulation, justifying the "Enhanced" classification in the secondary PTP framework.

9.2 Post-Bariatric Gut Hormone Remodeling

According to PubMed, RYGB induces a sustained enhancement of postprandial PYY and GLP-1 responses over at least 24 months (Pournaras et al., 2010). Simultaneously, ghrelin suppression following meals is normalized after RYGB, returning from a blunted (obesity-associated) pattern to a reference-concordant postprandial suppression profile (Engström et al., 2007; Samat et al., 2013). The dissociation between GIP (which shows variable or unchanged responses post-RYGB) and GLP-1 (which shows robust amplification) underscores the importance of analyte-specific classification rather than a single composite incretin score (Bojsen-Møller et al., 2015; Shin et al., 2010).

9.3 Functional Data Analysis in Periprandial Endocrinology

Functional Principal Component Analysis provides a principled decomposition of periprandial time-series into orthogonal modes of variation. PC1 captures global amplitude (the overall magnitude of the response), PC2 captures early-versus-late redistribution (reflecting, for example, the balance between first-phase and second-phase insulin secretion), and PC3 captures higher-order shape features such as biphasic patterns (Ramsay & Silverman, 2005; Yao et al., 2005). Sparse FPCA methods allow application to irregularly sampled periprandial data typical of clinical studies (Yao et al., 2005). The use of standardized z-scores within the classification vector Z_{ih} ensures comparability across analytes with different measurement scales and physiological ranges (Wang et al., 2016).

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